

# Chapter 21 Coulomb's Law

## Electric Charge

### Properties of Electric Charges

Two types of charges: positive and negative.

Electrostatics: The study of the interactions between charged particles **at rest**.

Matter is composed of elementary particles and atoms consist of **three fundamental particles**: electrons(nature's basic carrier of **negative charge**, can be gained or lost), protons(nature's basic carrier of **positive charge**,be held firmly in nucleus) and neutrons. Electric charge is a property of these particles

### Two Properties of Electric Charges

- Electric charge is conserved.
- Electric charge is quantized.

A fundamental unit of charge is  $e$ ,  $e = 1.6 \times 10^{-19}\text{C}$ . One electron has a charge of  $-e$  while one proton has a charge of  $+e$ .

## Conductors, Insulators and Induced Charges

### Definitions

Conductors: Materials that the **electric charges move freely** by an electric force.

Insulators: Materials that **electric charges do not move freely**.

Semiconductors: The conductivity of semiconductors are between those of insulators and conductors.

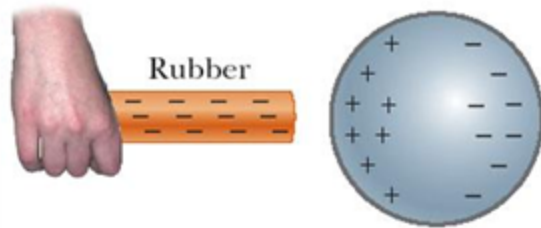
### Characteristics

| Type          | Conductivity | Resistivity | Forbidden Energy Gap | Number of free electrons | Examples           |
|---------------|--------------|-------------|----------------------|--------------------------|--------------------|
| Insulator     | Low          | High        | Wide                 | Small                    | paper, mica, glass |
| Semiconductor | Moderate     | Moderate    | Narrow               | Moderate                 | silicon, germanium |
| Conductor     | High         | Low         | No                   | Large                    | metals             |

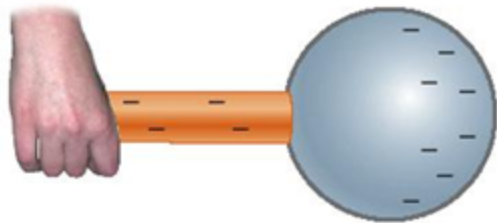
# Conduction Charging and Induction Charging

Two approaches to make a conductor becomes charged.

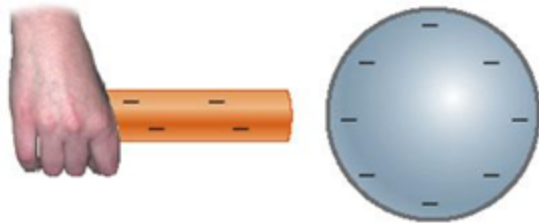
## Conduction Charging



(a) Before



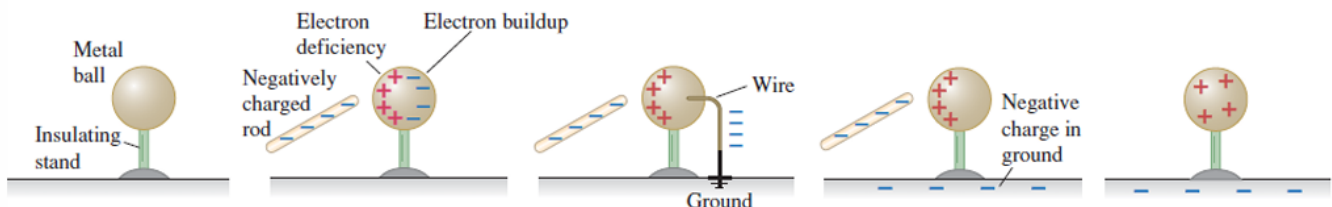
(b) Contact



(c) After breaking contact

- A charged rod is placed in contact with the sphere
- Some electrons on the rod can move to the sphere
- When the rod is removed, the sphere is left with negative charges
- The sphere being charged is always left with a charge having **the same sign** as the rod doing the charging

## Induction Charging



- A uncharged metal ball is placed on an insulating stand
- Use a charged rod to approach the metal ball. Negative charge on rod **repels** electrons, creating zones of negative and positive **induced charge**.
- Wire lets electrons build-up flow into ground.
- Wire removed and ball has only electron-deficient region of positive charge. Rod removed and the situation remains. Have the **different sign** charge.

## Polarization Effect

In general, in most **neutral** particles, the center of positive charge coincides with the negative. When a **charged object** appears, these centers may **separate slightly**. This realignment of charge on the surface of an **insulator** is known as **polarization**.

## Coulomb's Law

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{|q_1 q_2|}{r^2}$$

where  $\epsilon_0 = 8.854 \times 10^{-12} C^2 / N \cdot m^2$ ,  $k = \frac{1}{4\pi\epsilon_0} \approx 9.0 \times 10^9 N \cdot m^2 / C^2$  (called Coulomb Constant),  $F$  is the magnitude of interactions between charged particles.

It is attractive if the charges are of opposite signs and repulsive if the charges have the same signs. Resultant force equals to the **vector sum** of the forces.

Only to **point charges**.